

mathematics

UNIT I – Differential Equations of First Order

- Exact differential equations
 - Integrating factors
 - Linear differential equations
 - Bernoulli's differential equations
 - Riccati's differential equations
 - Clairaut's differential equations
 - Orthogonal trajectories of a given family of curves
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UNIT II – Differential Equations of Higher Order

- Solution of second and higher order linear homogeneous differential equations with constant coefficients
 - Reduction of order method:
 - Linear homogeneous second-order differential equations with variable coefficients
 - Non-homogeneous linear differential equations
 - Method of variation of parameters
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- Solution of Euler–Cauchy differential equation
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UNIT III – Laplace Transforms

- Laplace transform
 - Inverse Laplace transform
 - Properties of Laplace transforms
 - Properties of inverse Laplace transforms
 - Convolution theorem (without proof)
 - Solution of ordinary differential equations using Laplace transforms
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UNIT IV – Numerical Methods – I

Solution of Algebraic Equations

- Polynomial equations
- Transcendental equations
- Bisection method
- Iteration method
- Newton–Raphson method
- Regula–Falsi method

Finite Differences & Interpolation

- Finite differences:
 - Forward differences
 - Backward differences
 - Central differences
- Symbolic relations and separation of symbols
- Interpolation methods:
 - Newton's forward interpolation formula
 - Newton's backward interpolation formula
 - Lagrange's interpolation method

- Euler's method
- Modified Euler's method
- Fourth-order Runge–Kutta method

UNIT V – Numerical Methods – II

Numerical Integration

- Trapezoidal rule
- Simpson's 1/3 rule
- Simpson's 3/8 rule

Numerical Solution of ODEs

- Taylor's series method
- Picard's method